

Sequence Optimization of Multispacecraft Multitarget Rendezvous Missions with a Coevolutionary Algorithm*

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ABSTRACT

The optimization of multispacecraft multitarget rendezvous sequences is a critical component of mission planning for endeavors such as space debris removal and multiasteroid exploration. This approach has significant implications for improving mission efficiency and resource utilization. In this paper, a coevolutionary algorithm that synergistically combines a genetic algorithm (GA) with ant colony optimization (ACO) is proposed to solve a multispacecraft multitarget rendezvous sequence planning problem. The proposed algorithm employs a hierarchical strategy for target allocation and sequence planning, seamlessly integrating these two layers into a unified coevolutionary framework. To address the time-dependent challenges in space target rendezvous sequence planning, the ACO algorithm incorporates a time-local search strategy to optimize the rendezvous sequence and rendezvous time. Two local search strategies, which are called tail addition and sequence replanning, are further proposed to address the challenge associated with remaining targets in fuel-constrained missions. The experimental results highlight the effectiveness of the coevolutionary algorithm in solving the multispacecraft multitarget rendezvous sequence planning problem, with performance improvements of 8.4–10.2% in optimality over traditional methods. In the modified scenarios of the 12th Global Trajectory Optimization Competition, the algorithm's optimization results are superior to those of the winning team, highlighting its exceptional optimization capabilities.

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CCS CONCEPTS

• Applied computing → Physical sciences and engineering → Aerospace

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Multispacecraft multitarget rendezvous; Target allocation; Space mission planning; Evolutionary algorithm; GTOC

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1 Introduction

Space target rendezvous missions are critical and widely conducted space operations, encompassing a diverse range of objectives, such as debris removal [1]–[3], on-orbit servicing [4]–[6], and deep space exploration [7]–[9]. Numerous space agencies have devised and executed series of such missions, achieving significant advancements. For example, the ClearSpace-1 mission [10], a joint initiative between the European Space Agency and ClearSpace, was an initial attempt at orbital debris removal. Similarly, Northrop Grumman successfully demonstrated satellite life-extension capabilities through Mission Extension Vehicle-1 (MEV-1) and Mission Extension Vehicle-2 (MEV-2) [4], whereas the Hayabusa2 mission [11, 12] made groundbreaking progress in the exploration of multiple asteroids. Collectively, these missions mark substantial advancements in the domains of space debris removal, satellite life-extension, and deep space exploration.

Although most existing missions were tailored primarily for single-target or small-scale target scenarios, the growing complexity and scale of future space rendezvous operations render

it increasingly challenging for a single spacecraft to fulfill mission requirements. Consequently, the adoption of multispacecraft cooperation has emerged as an advantageous strategy. This shift has prompted numerous researchers to explore the more complex and demanding problem of multispacecraft multitarget rendezvous sequence planning (MMRSPP). Fundamentally, the MMRSPP is similar to the classic multiple traveling salesman problem (mTSP) [13], as both involve multiple agents (salesmen or spacecraft) tasked with visiting multiple targets (cities or space targets). However, the distinction lies in the fact that the MMRSPP entails the optimization of continuous variables. Owing to the time-dependent characteristics of space targets, it is necessary to include rendezvous times as optimization variables. Additionally, the orbital dynamics of space targets introduce further complexity. Even with fixed rendezvous targets and times, the fuel required for transfers depends on the precise transfer trajectory of the spacecraft. Consequently, optimizing the spacecraft trajectory for each transfer arc becomes indispensable. This makes the MMRSPP a multilayered and nested optimization problem involving both continuous and discrete variables.

For the MMRSPP, common solution approaches include iterative planning or simultaneous planning of multiple rendezvous sequences, as well as truncating a long sequence to obtain multispacecraft rendezvous sequences. Luo et al. [14] proposed a grouped ant colony algorithm to address this problem, iteratively employing the algorithm to construct sequences until the fuel limit of the current spacecraft is exceeded, thereby producing multiple rendezvous sequences. Similarly, Zhang et al. [15] introduced a multitree search algorithm designed for the MMRSPP; it generates a comprehensive solution by expanding multiple tree branches. Huang et al. [16] utilized a genetic algorithm to optimize entire serial rendezvous sequences, segmenting them on the basis of spacecraft resource constraints to derive multiple rendezvous sequences. In addition, some researchers have explored preplanning approaches to establish a database of rendezvous sequences combined with optimization methods to obtain sequence solutions that meet certain constraints. Zhang et al. [17] generated a database of single rendezvous sequences via tree searches and subsequently optimized the combination scheme via a genetic algorithm. Other studies have approached this problem through model transformation and traditional mathematical programming techniques. Bang et al. [18] decomposed the issue into a nonlinear programming problem and an integer linear programming (ILP) problem, resolving the ILP component with a column generation algorithm. Another effective strategy for addressing the MMRSPP is to deconstruct the problem into two hierarchical levels: target allocation and sequence planning. This hierarchical framework allows for the joint resolution of the problem. Viewed through this lens, the MMRSPP integrates elements of the knapsack problem and the traveling salesman problem (TSP) [19, 20] while also accounting for time-dependent variables. Chen et al. [21] employed the *k*-medoid clustering method to group targets that are easily transferable, followed by sequence planning via a tree search method. However, it is worth noting that the allocation scheme in

reference [21] did not evolve dynamically. Furthermore, spacecraft were assumed to have sufficient fuel and time resources in reference [21], meaning that the proposed solution method might face challenges in scenarios where resources are limited, potentially resulting in the inability to rendezvous with all targets.

To address the aforementioned challenges, a coevolutionary algorithm (CEA) that integrates a genetic algorithm (GA) and an ant colony optimization (ACO) algorithm is proposed in this study to solve the MMRSPP. By decomposing the task of constructing multiple sequences into two hierarchical levels—target allocation and single-sequence construction—the framework leverages the GA for high-level target allocation and the ACO algorithm for sequence construction at lower levels. Within this framework, individuals are encoded and decoded via the GA, whereas rendezvous sequences are constructed via the ACO algorithm. The ACO algorithm incorporates a time-local optimization strategy, which explicitly considers the influence of the time dimension on the rendezvous sequences. Additionally, two customized local optimization strategies are proposed to further refine the rendezvous sequences for multiple spacecraft, aiming to maximize mission benefits given resource constraints. The final optimized planning results are iteratively fed back to both the GA individuals and the ACO algorithm individuals, facilitating the generational evolution of genes and pheromones. These updates are subsequently reflected in the target allocation and sequence planning schemes for the next-generation population. Experimental validation is conducted via modified scenarios derived from the 12th edition of the Global Trajectory Optimization Competition (GTOC12). The CEA outperforms the winning solution from the JPL, demonstrating its superior effectiveness in addressing the MMRSPP.

The remainder of this paper is structured as follows. Section 2 outlines the problem description, and the optimization model is formulated. In Section 3, the detailed workflow of the CEA is explored, and its key steps are introduced. Section 4 presents the simulation experiments and the corresponding results. Finally, Section 5 concludes the paper with a concise summary.

2 Problem Description and Optimization Model

Multispacecraft multitarget rendezvous missions necessitate the coordinated efforts of several spacecraft to achieve successful rendezvous with all designated targets while adhering to mission constraints, as depicted in Figure 1. Throughout the mission, each spacecraft is assigned a set of targets with which to rendezvous. Upon completing a rendezvous with one target, the spacecraft promptly travels to the next target, continuing this process until all the targets have been successfully reached. Planning rendezvous sequences requires comprehensive consideration of various resource constraints, such as fuel consumption and time constraints, in conjunction with task-specific constraints. The objective is to determine the optimal rendezvous sequence, rendezvous time, and transfer trajectories for each spacecraft, thereby maximizing the overall mission benefit.

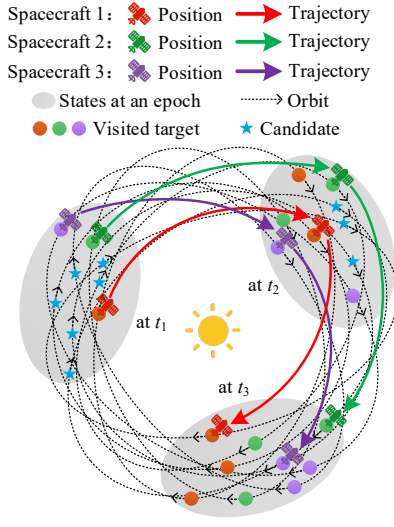


Figure 1: A schematic diagram of the multispacecraft multitarget rendezvous mission scenario

In a mission scenario involving N spacecraft $\{s_1, s_2, \dots, s_N\}$ equipped with electric propulsion systems to rendezvous with n ($n \geq N$) targets $\{a_1, a_2, \dots, a_n\}$, the design variables for the optimization model of the MMRSP can be defined as follows:

$$X = (a_{i,j}, t_{i,j}) \quad i = 1, 2, \dots, n; j = 1, 2, \dots, n_i \quad (1)$$

where $a_{i,j}$ represents the j -th rendezvous target of s_i , $t_{i,j}$ represents the time at which s_i rendezvous with $a_{i,j}$, and n_i denotes the total number of targets assigned to s_i .

The objective function is typically established to minimize fuel consumption, reduce total mission time, or maximize mission benefits. In this work, fuel consumption and mission time are treated as constraints, whereas maximizing mission benefits is established as the primary objective. Specifically, the mission benefit in this paper is defined as a function of the rendezvous time, as illustrated below:

$$\max \sum_{i=1}^N f_{t,i}(t_{i,1}, t_{i,2}, \dots, t_{i,n_i}) \quad (2)$$

where $f_{t,i}$ represents the benefit function of spacecraft s_i in relation to the rendezvous time $(t_{i,1}, t_{i,2}, \dots, t_{i,n_i})$.

Considering a multispacecraft multitarget rendezvous mission with limited fuel and time, the constraints can be formulated as follows:

$$\begin{cases} \sum_{i=1}^N n_i = N \\ a_{i,j_1} \neq a_{i,j_2} & \text{if } i_1 \neq i_2 \text{ or } j_1 \neq j_2 \\ t_0 \leq t_{i,j} < t_{i,k} \leq t_f & \text{if } j < k \\ \sum_{j=1}^{n_i-1} g_m(a_{i,j}, a_{i,j+1}, t_{i,j}, t_{i,j+1}, m_{i,j}) + \Delta m_{e2a_{n_i}} \leq M_{fuel,i} \end{cases} \quad i = 1, 2, \dots, N \quad (3)$$

Constraints 1 and 2 ensure that each target has rendezvoused exactly once. Constraint 3 ensures the monotonicity of the

rendezvous times and that they lie within the mission time window. Constraint 4 ensures that the fuel consumption of each spacecraft does not exceed the fuel mass it carries, where $M_{fuel,i}$ represents the fuel mass carried by the spacecraft, g_m represents the fuel consumption calculation function, and $\Delta m_{e2a_{n_i}}$ represents the fuel consumption of s_i returning to Earth from a_{n_i} . Given the transfer arc (including the initial and final targets $a_{i,j}$ and $a_{i,j+1}$ and the initial and final times $t_{i,j}$ and $t_{i,j+1}$) and the initial spacecraft mass $m_{i,j}$, the fuel consumption mass can be estimated via the empirical formula proposed in reference [22]. The formula for $m_{i,j}$ is generally as follows:

$$m_{i,j+1} = m_{i,j} - g_m(a_{i,j}, a_{i,j+1}, t_{i,j}, t_{i,j+1}, m_{i,j}) + \Delta m_{add} \quad (4)$$

where Δm_{add} represents the additional mass change associated with specific rendezvous missions, such as the deployment of equipment or the collection of resources.

3 A Coevolutionary Algorithm for Solving the MMRSP

We decompose the MMRSP into a two-stage optimization problem. In the first stage, the proposed CEA is employed to optimize the rendezvous paths (rendezvous sequences and rendezvous times) for multiple spacecraft. In the second stage, the rendezvous sequences of each spacecraft are fixed, and an improved DE algorithm [23] is employed to optimize the rendezvous times for each spacecraft. When individual fitness values are calculated, the indirect method proposed in reference [24] is used to optimize the minimum-fuel low-thrust trajectories of the spacecraft, aiming to maximize the objective function value under the relevant fuel constraints (ensuring that the remaining fuel mass is greater than but close to zero). This paper focuses on the CEA in the first stage, whereas the optimization process in the second stage was explored in reference [22].

3.1 Interactive Evolutionary Process of the Coevolutionary Algorithm

Figure 2 illustrates the hierarchical computation and interactive evolutionary process of the coevolutionary algorithm, encompassing the workflows of target allocation, sequence planning, and local optimization. Additionally, the mutual interplay among the optimization results, algorithm parameters, and operators is shown. Within the CEA, the GA framework predominantly facilitates target allocation. By employing evolutionary operations such as selection, crossover, and mutation, the target allocation schemes are iteratively refined to achieve optimal evolution at the allocation level. Moreover, the ACO algorithm framework is primarily used to optimize rendezvous paths on the basis of the given allocation results. By integrating local time optimization and fuel estimation methods, the optimal rendezvous sequence and time for each individual can be generated. A detailed explanation of the algorithm's key processes is provided in Sections 3.2 to 3.6.

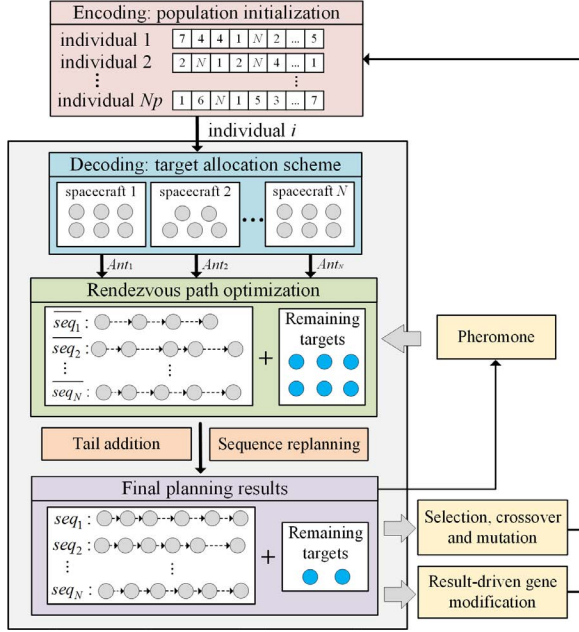


Figure 2: Computational and evolutionary process of the CEA

3.2 Encoding and Decoding

Given a target set $\{a_1, a_2, \dots, a_n\}$, the chromosome of each individual is encoded via an integer encoding method ($G_1, G_2, \dots, G_n$), where $G_i = 1, 2, \dots, N$ indicates that target a_i is assigned to spacecraft s_{G_i} . Once the individual encoding is completed, the chromosome genes are traversed to decode, as shown in Figure 3, where A_i is the set of targets assigned to spacecraft s_i . This straightforward encoding method ensures that each chromosome corresponds to a uniquely determined allocation scheme, devoid of any information regarding the rendezvous sequence or time. This specifically enhances the search and evolution of allocation schemes. During population initialization, the genes G_i associated with chromosome are randomly assigned equal probabilities from $\{1, 2, \dots, N\}$, ultimately generating N_p individuals to constitute the initial population.

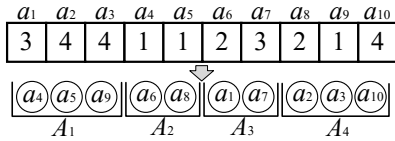


Figure 3: Schematic diagram of encoding and decoding

3.3 Rendezvous Path Optimization Based on the ACO Algorithm

In the CEA, the rendezvous paths are optimized through the solution construction method of the ACO algorithm combined with a time-local optimization method (TLOACO) [22]. Given the departure target a_j , arrival target a_k , departure time $T_{i,j}^d$, and spacecraft mass $m_{i,j}$ at departure, the time-local optimization method proposed in reference [22] yields the corresponding

transfer time $\Delta t_{i,j,k}$ (measured in days), with Δt as the step size. Consequently, the solution construction process for the rendezvous path of spacecraft s_i is as follows.

1. Initialize the candidate target set $Z = A_i$. Traverse the candidate target a_k from the Z , calculate the transfer time $\Delta t_{i,j,k}$, and determine its selection probability:

$$p_{i,j}(k) = \begin{cases} \frac{\tau_{jk}^\alpha (\gamma \Delta t_{i,j,k} + \Delta \tilde{m}_{i,j,k})^\beta}{\sum_{z \in Z} \tau_{jz}^\alpha (\gamma \Delta t_{i,j,z} + \Delta \tilde{m}_{i,j,z})^\beta} & \Delta \tilde{m}_{i,j,k} \leq \Delta \tilde{m}_{\max}, k \in Z \\ 0 & \text{else} \end{cases} \quad (5)$$

Here, $\Delta \tilde{m}_{i,j,z}$ represents the estimated fuel consumption for the i -th spacecraft to transfer from target a_j to target a_z , and α , β , and γ are weighting parameters within the range $[0, 1]$. $\Delta \tilde{m}_{\max}$ is the larger of $M_{fuel,i}^{rem}$ and $\Delta m_{\max}$, where $M_{fuel,i}^{rem}$ denotes the remaining fuel mass of the i -th spacecraft in kilograms, and $\Delta m_{\max}$ is the maximum fuel consumption limit for a single transfer.

2. Select the next rendezvous target based on either the maximum probability or roulette wheel selection method, with equal probability for each method, and then remove the selected target from Z .
3. Repeat steps 1 to 2 until either the time or fuel constraints in Equation (3) are no longer satisfied or the number of candidate targets is reduced to zero.
4. Evaluate the objective function value $f_{i,i}(t_{i,1}, t_{i,2}, \dots, t_{i,n_i})$.
5. Repeat steps 1 to 4 a total of n_s to determine the optimal sequences $\overline{seq}_i = (\overline{a_{i,1}}, \overline{a_{i,2}}, \dots, \overline{a_{i,n_i}})$ and rendezvous times $\overline{T}_i = (\overline{t_{i,1}}, \overline{t_{i,2}}, \dots, \overline{t_{i,n_i}})$. Variables with an overline represent intermediate process variables rather than the final planning results.

The rendezvous paths for N spacecraft are determined through the aforementioned process, and the fitness value of each individual is subsequently evaluated via Equation (2). The optimization of the rendezvous paths is not based on the complete ACO algorithm but rather on the iterative application of the solution construction method.

3.4 Local Optimization of Rendezvous Paths

Owing to constraints associated with target allocation schemes, spacecraft fuel limitations, or mission duration (typically related to the allocation scheme), the sequence planning process described in Section 3.3 may lead to certain targets not being successfully rendezvoused. These mission targets are referred to as remaining targets and together constitute the set of remaining targets $A_{r,0}$. In this section, two local optimization methods for rendezvous with the remaining targets are proposed.

3.4.1 Tail addition. The tail addition strategy (TAS) is used to continuously identify the spacecraft with the most remaining fuel and probabilistically select targets from $A_{r,0}$, adding them to the end of the rendezvous sequence. This procedure continues until either all spacecraft are devoid of sufficient fuel to rendezvous with any remaining targets or until $A_{r,0}$ is empty. The pseudocode

for the TAS is as follows, where $\overline{Seq} = \{\overline{seq_1}, \overline{seq_2}, \dots, \overline{seq_N}\}$ and $\overline{T_{set}} = \{\overline{T_1}, \overline{T_2}, \dots, \overline{T_N}\}$. The strategy is executed n_e times to achieve sufficient local optimization results.

Algorithm 1: Tail Addition

Input: An individual, rendezvous sequences $\overline{Seq}$, rendezvous times $\overline{T_{set}}$, the corresponding remaining targets $A_{r,0}$ and other information

```

1: while True do
2:   for each spacecraft  $s_i$  in  $S = \{s_1, s_2, \dots, s_N\}$  do
3:      $p(i) = M_{fuel,i}^{rem} / \sum_{j=1}^{card(S)} M_{fuel,j}^{rem}$ 
4:   end for
5:   Select a spacecraft  $s_i$  based on probability
6:   for each candidate target  $z_i$  in  $Z = A_{r,0}$  do
7:     Calculate the selection probability  $p(z_i)$  according
       to Equation (5)
8:   end for
9:   if  $\forall z_i \in Z, p(z_i) = 0$  then
10:    Update  $S = S \setminus \{s_i\}$ 
11:   else
12:    Select a target  $z_i$ , and update  $Z = Z \setminus \{z_i\}$ 
13:   end if
14:   if  $Z = \emptyset$  or  $S = \emptyset$  then
15:     break
16:   end while
17: Calculate the fitness value according to Equation (2)

```

Output: Updated $\overline{Seq}$ and $\overline{T_{set}}$ for individuals as well as the corresponding remaining targets $A_{r,1}$ and other information.

3.4.2 Sequence replanning. The sequence replanning strategy (SRS) combines the existing rendezvous target set and the remaining target set to generate a new rendezvous sequence and rendezvous time. The pseudocode for the SRS is presented below. If $A_{r,1} \neq \emptyset$, the SRS will be executed n_r times to achieve locally optimized results.

Algorithm 2: Sequence Replanning

Input: An individual, $\overline{Seq}$ and $\overline{T_{set}}$, the corresponding remaining targets $A_{r,1}$ and other information

```

1: Generate a random permutation  $(p_1, p_2, \dots, p_N)$  of  $\{1, 2, \dots, N\}$ .
2: for  $i = 1$  to  $N$  do
3:    $\overline{A}_{p_i} = \overline{A}_{p_i} \cup A_{r,1}$ , where  $\overline{A}_{p_i} = \{\overline{seq_{p_i}}[1], \overline{seq_{p_i}}[2], \dots\}$ 
4:   Plan the rendezvous path according to the method
     described in Section 3.3
5:   Calculate the fitness value  $f_{t,p_i}(t_{p_i,1}, t_{p_i,2}, \dots)$ 
6:   if  $f_{t,p_i}(t_{p_i,1}, t_{p_i,2}, \dots) > f_{t,p_i}(t_{p_i,1}, t_{p_i,2}, \dots)_{original}$  then
        $f_{t,p_i}(t_{p_i,1}, t_{p_i,2}, \dots)_{original} = f_{t,p_i}(t_{p_i,1}, t_{p_i,2}, \dots)$ 

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7:   Update  $\overline{seq_{p_i}}$ ,  $\overline{T_{p_i}}$  and  $A_{r,1}$ 
8:   end if
9:   if  $A_{r,1} = \emptyset$  then
10:    break
11:   end for
12: Calculate the fitness value according to Equation (2)
Output: Updated  $\overline{Seq} = \{\overline{seq_1}, \overline{seq_2}, \dots, \overline{seq_N}\}$  and
 $\overline{T_{set}} = \{\overline{T_1}, \overline{T_2}, \dots, \overline{T_i}\}$  for individuals as well as the
corresponding remaining targets  $A_{r,2}$  and other information

```

3.5 Result-Driven Individual Gene Modification

Following the global and local optimization methods described in Sections 3.2–3.4, the rendezvous sequences Seq and rendezvous times T_{set} for the N spacecraft associated with an individual and the final remaining targets $A_{r,2}$, are obtained. However, owing to the effects of local optimization, the rendezvous sequence of a spacecraft seq_i may include targets from the remaining target set $A_{r,0}$ or $A_{r,1}$. Consequently, the allocation of such targets may change, leading to a discrepancy between the initial chromosome encoding result $(G_1, G_2, \dots, G_n)$ and the final rendezvous sequence result Seq . To address this issue, the genes associated with these targets require adaptive mutation, and this operation is referred to as result-driven individual gene modification. The mutation method involves traversing to find the index k of $a_{i,j}$ in $(a_1, a_2, \dots, a_n)$. If $i \neq G_k$, then G_k is updated to i . Ultimately, the final chromosome encoding results $(G_1, G_2, \dots, G_n)$ are obtained.

3.6 Selection, Crossover, Mutation, and Pheromone Updating

The selection operation is based on the roulette wheel method. The crossover operation uses modified uniform crossover. Specifically, given the chromosome encoding result $(G_1, G_2, \dots, G_n)_{final,1}$, the final remaining targets $A_{r,2}^1$ of parent individual 1, the chromosome encoding result $(G_1, G_2, \dots, G_n)_{final,2}$ and the final remaining targets $A_{r,2}^2$ of parent individual 2, $i = 1, 2, \dots, n$ is traversed. If $a_i \in A_{r,2}^1 \cap A_{r,2}^2$, then the crossover probability P_c of G_i is 0.9; if $a_i \in (A_{r,2}^1 \cup A_{r,2}^2) \setminus (A_{r,2}^1 \cap A_{r,2}^2)$, then $P_c = 0.7$; and if $a_i \notin A_{r,2}^1 \cup A_{r,2}^2$, then $P_c = 0.1$. The mutation operation involves modified uniform mutation. Given the chromosome encoding result $(G_1, G_2, \dots, G_n)_{final}$ and the final remaining targets $A_{r,2}$, $i = 1, 2, \dots, n$ is traversed. If $a_i \in A_{r,2}$ (indicating its initial allocation scheme is inappropriate), the mutation probability P_m of G_i is set to 0.9; otherwise, $P_m = 0.1$. When generating a new population, the top 30% of individuals from the previous generation are directly retained in the new generation.

The initial pheromone value τ_0 is set to 1 when $g = 1$. For $g = 2$, it is updated to $1/L_{best,1}$, where $L_{best,1}$ represents the objective function value of the best individual in the first generation. Here, g represents the current iteration generation. Pheromone updates consist of classic local and global pheromone updates. The above

parameters were chosen based on empirical findings and experimental results, without hyperparameter optimization, which can be improved in future work.

3.7 Pseudocode for the Coevolutionary Algorithm

The pseudocode for the CEA is as follows, where $G_{\max}$ represents the maximum number of iterations.

Algorithm 3: Coevolutionary Algorithm

Input: Initial states of n mission targets and N spacecraft

- 1: Initialize the basic parameters of the algorithm and the population based on the method in section 3.2.
- 2: **for** $g = 1 : G_{\max}$ **do**
- 3: **for** $p = 1 : N_p$ **do**
- 4: **for** each spacecraft **do**
- 5: Use the method in section 3.3 for sequence planning
- 6: **end for**
- 7: Perform local optimization using the methods in section 3.4
- 8: Perform result-driven individual genetic modification based on the method described in section 3.5
- 9: Perform local pheromone update according to the method in section 3.6
- 10: **end for**
- 11: Update the optimal individual
- 12: Perform global pheromone updating based on the method in section 3.6
- 13: Use the methods in section 3.6 for selection, crossover and mutation
- 14: **end for**

Output: The rendezvous path of N spacecraft

4 Simulation Experiment

This research builds upon the background of GTOC12 [25], a multispacecraft multitarget rendezvous mission scenario is constructed under fuel constraints. Notably, the mission analyzed in the case study is not the original mission. Each spacecraft is equipped with an electric propulsion system with a specific impulse of $I_{sp} = 4000$ s and a maximum thrust of $T_{\max} = 0.6$ N. The spacecraft are required to rendezvous with n mission targets to collect resources, where the initial states of N spacecraft and n mission targets are predefined. The total mission benefit is proportional to the mass of the collected resources, as defined by the following equation:

$$J = \sum_{i=1}^n M_{res,i} = \sum_{i=1}^n k_m (t_i - t_{i,0}) \quad (6)$$

Here, $M_{res,i}$ is the mass of resources collected from the i -th target, which is proportional to the duration of resource extraction, measured in kilograms, and corresponds to the parameter Δm_{add} in Equation (4). t_i denotes the time at which the spacecraft

rendezvous with the i -th target to collect resources, whereas $t_{i,0}$ represents the known time when the resource extraction device is deployed at the i -th target. The proportionality coefficient k_m is set to 10 kg/yr. The specific spacecraft and mission targets correspond to spacecrafts 6-19 and the 130 mission targets they visit, as outlined in the solutions of the champion team JPL. Further details are presented in Table 4. Based on experimental results and practical experience, the basic parameters of the CEA are presented in Table 1.

Table 1: Basic parameters of the CEA

Parameter	Value	Parameter	Value
$G_{\max}$	100	ρ	0.1
N_p	30	ζ	0.1
γ	3/7	n_s	30
α	1	n_e	30
β	5	n_r	10

4.1 Parameter Configuration

To select the iteration step size Δt for the time-local optimization method used in the algorithm, 50 experiments are conducted for four different values. The statistical optimization results are presented in Table 2, where T_E is the execution time. Ultimately, the time iteration step Δt for the time-local optimization algorithm nested within the ACO algorithm is determined to be 30 days.

Table 2: Statistical optimization results corresponding to different Δt values

Index	Δt (day)	J /(kg)	$\text{card}(A_{r,2})$	T_E (s)
1	20	9711.54	7.56	812.5
2	30	9877.65	0.84	494.8
3	40	9690.03	0.72	419.7
4	50	9507.03	0.64	422.1

4.2 Performance of the Algorithm

Two commonly used methods for MMRSP are employed to assess the results and solution efficiency of the CEA: the grouped ant colony optimization (GACO) algorithm [26] and the hybrid algorithm (genetic algorithm + greedy algorithm, GGA). For both GACO and the GGA, the number of evolutionary generations is set to 100, and the population size is set to 300. Each of the three algorithms is independently executed 100 times. The results are presented in Table 3, which lists the objective function values J , the number of remaining targets $\text{card}(A_{r,2})$ and the execution time T_E .

The statistical results presented in Table 3 indicate that the CEA performs best for both J and $\text{card}(A_{r,2})$. Specifically, the average objective function value of the CEA is 9874.9 kg, with this approach significantly outperforming GACO (8964.2 kg) and the GGA (9113.5 kg). The average J obtained by the CEA is 8.4% to 10.2% higher than that of the other two methods, demonstrating the superiority of the CEA. Additionally, the CEA yields an average of 0.9 remaining targets, compared with 8.5 and 13.9 for

GACO and the GGA, respectively, underscoring a notable advantage in terms of $\text{card}(A_{r,2})$. Although the CEA requires more time to solve the problem, its superior performance justifies the additional computational cost. The CEA employs a GA framework and an ACO algorithm to independently address allocation and sequence-level challenges, facilitating collaborative evolution. Consequently, the CEA delivers superior optimization performance compared to the other methods.

Table 3: Optimization results of the three algorithms

Parameter	Algorithm	Min.	Max.	Mean	Std.
J (kg)	CEA	9680.3	10036.0	9874.9	80.6
	GACO	8626.3	9212.9	8964.2	115.3
	GGA	8857.0	9320.4	9113.5	78.1
$\text{card}(A_{r,2})$	CEA	0	3	0.9	0.9
	GACO	5	14	8.5	1.7
	GGA	8	19	13.9	2.4
T_E (s)	CEA	428.6	511.7	477.4	19.2
	GACO	116.6	559.7	295.8	81.9
	GGA	140.2	236.5	203.6	11.1

Figure 4 presents the results of the three algorithms, including a histogram of the probability distributions of the objective function values and a scatter plot with three-dimensional data. The results achieved by the CEA exhibit a distinct advantage over those of the other two algorithms, displaying markedly superior performance for both J and $\text{card}(A_{r,2})$. This highlights the CEA's exceptional ability to maximize mission benefits.

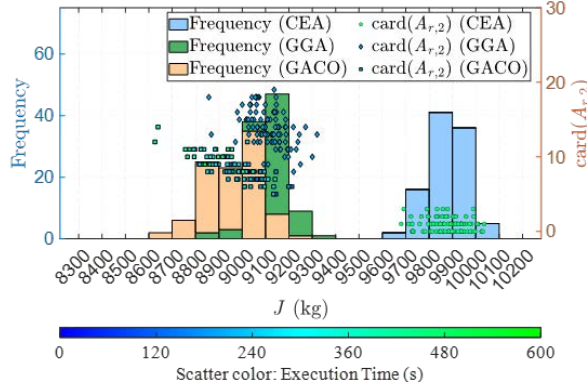


Figure 4: Visualization of the results of the three algorithms

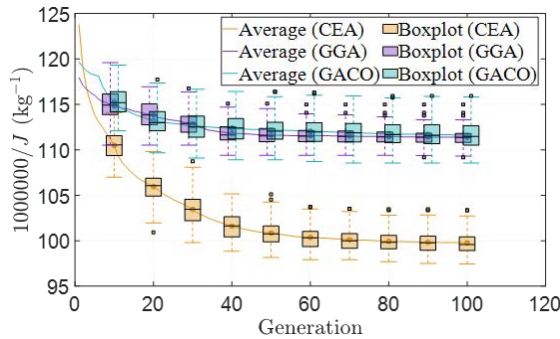


Figure 5: Average convergence curves of the three algorithms

Figure 5 shows the convergence curves of the average objective function values across 100 independent optimizations for the three algorithms, as well as box plots and scatter plots at 10-generation intervals. The convergence curve of the CEA is smooth, and the solution stability is high. Furthermore, the CEA box plots are narrower than those for the other methods and tend to progressively narrow and become more stable. While the sequence for the GGA with a greedy algorithm displays more consistent performance than that for the CEA, the quality of the CEA solution is markedly superior. In contrast, the GACO result exhibits significant fluctuations, along with poor convergence and stability.

4.3 Effectiveness of the Local Optimization Method

To validate the effectiveness of the local optimization strategy for improving J and reducing $\text{card}(A_{r,2})$, we compared the results under different combinations of local optimization strategies. Each configuration was independently executed 100 times, and the statistical results of the optimization are shown in Table 4. These results include the minimum, maximum, mean, and standard deviation of J , $\text{card}(A_{r,2})$ and T_E . Furthermore, 30 random data points are selected to visually compare the results, as illustrated in Figure 6.

Table 4: Optimization results under different local optimization strategies

Parameter	Strategy	Min.	Max.	Mean	Std.
J (kg)	TAS+SRS	9645.5	10149.2	9895.5	87.6
	TAS	8881.7	9752.2	9476.3	139.0
	/	8469.4	9523.6	9054.7	240.2
$\text{card}(A_{r,2})$	TAS+SRS	0	3	0.7	0.7
	TAS	3	13	6.9	1.9
	/	6	21	13.4	3.2
T_E (s)	TAS+SRS	237.5	529.9	470.1	47.4
	TAS	135.0	172.1	155.5	7.4
	/	15.5	74.3	65.6	11.7

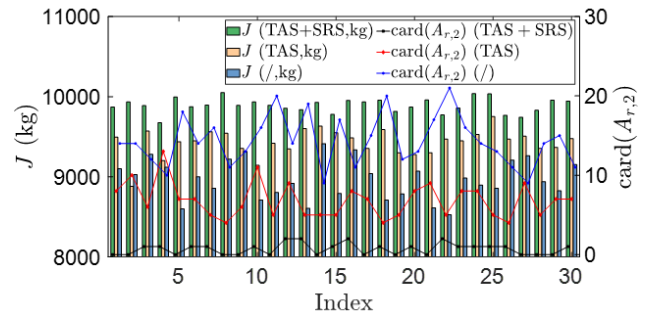


Figure 6: Comparison of 30 optimization results for the three algorithms

The experimental results show that TAS+SRS outperforms the other strategies for both J and $\text{card}(A_{r,2})$. The mean objective function value reaches 9895.5 kg, which is markedly greater than

that of the other strategies, and the average number of remaining targets is reduced to just 0.7. Notably, this strategy demonstrates minimal variability, with a standard deviation of only 87.6 for J , indicating exceptional stability. Particularly in application scenarios where high-quality solutions are critical, TAS+SRS emerges as the optimal choice. These findings underscore the effectiveness and critical role of local optimization strategies in addressing the MMRSP.

4.4 Final Results

The multispacecraft multitarget rendezvous mission detailed in section 4.1 is planned via the proposed solution framework, with approximately 500 independent runs conducted in the first stage. The final results are shown in Table 5, including the rendezvous sequence for each spacecraft and the objective function value. The

asteroids listed in the table are numbered according to the dataset of 60,000 asteroids provided in reference [25]. Although the MMRSP addressed in this study is modeled with certain modifications on the basis of the competition problem, the objective function values are the same. This enables a direct comparison to highlight the effectiveness of the algorithm and solution framework. The objective function value of the original solution is 10,290.1 kg, whereas the solution of the new objective function is 10,383.0 kg, surpassing the winning solution by 92.9 kg. This means that with the same fuel resources, the mined mass per spacecraft increases by an average of 6.6 kg, and the average rendezvous time for each target is delayed by 26.1 days. The superiority of the final results further demonstrates the effectiveness of the proposed algorithm in solving the MMRSP.

Table 5: Final optimization results, including the rendezvous sequences of 14 spacecraft

Spacecraft	Rendezvous sequences (asteroid number)	J (kg)
1	20594, 47630, 37322, 3241, 23056, 3896, 5441, 10177, 19906	793.95
2	28483, 29031, 29154, 50586, 10785, 22459, 7976, 56866, 43207, 12029	757.27
3	23054, 28090, 50972, 20959, 13487, 26188, 46460, 30029, 53706, 4210	872.57
4	14485, 25102, 35812, 34927, 40804, 47691, 5095, 57189, 26632	763.82
5	59481, 21677, 42983, 46418, 9825, 24020, 24025, 25279, 27464, 36243	736.55
6	33571, 43326, 15728, 51591, 37117, 3750, 47047, 46990	576.52
7	57844, 57913, 20732, 3302, 19589, 53223	550.84
8	28604, 56502, 7352, 51417, 17697, 9070, 32088, 23987, 46751, 15184	817.32
9	12628, 2311, 49915, 51614, 9260, 2434, 47817, 27751, 6856, 4367, 36210	853.69
10	4812, 13655, 23483, 33878, 288, 27262, 29656, 53592, 23795, 59262, 8064	842.58
11	53400, 5670, 21105, 1574, 13688, 51159, 12402, 22760	679.22
12	14266, 6737, 48748, 6405, 53908, 19702, 13467, 39557, 12707, 28884	742.53
13	36091, 3196, 26584, 11721, 5085, 5707, 15711, 44766, 37818	704.57
14	14516, 29390, 53074, 38744, 14723, 16356, 23438, 46343, 58468	691.63
Final solution		10383.0

5 Conclusions

In this paper, a coevolutionary algorithm for the MMRSP is proposed. The algorithm integrates a GA with ACO to form a hierarchical coevolutionary framework. Additionally, two customized local optimization strategies are designed to further improve the mission benefits. First, the GA is applied to optimize target allocation, and the target allocation results are encoded as genes for each individual. Through evolution operations such as selection, crossover, and mutation, the algorithm continuously refines the match between multiple spacecraft and targets, thereby generating high-quality inputs for sequence planning. Second, the solution construction method of the ACO algorithm combined with the time local optimization method effectively addresses the complex time-dependent characteristics of space targets and plays a key role in sequence planning. Finally, two local optimization strategies, tail addition and sequence replanning, are proposed to refine the rendezvous sequence. Furthermore, a result-driven gene modification operation is used to input the final sequence

planning results back into the target allocation layer, enabling the algorithm to adaptively adjust the target allocation scheme during iterative evolution. This process achieves effective coordination between global optimization and local optimization. On the basis of a modified GTOC12 case study, the results obtained with the proposed coevolutionary algorithm surpass those achieved by the competition-winning team (the JPL). This underscores the algorithm's effectiveness and superiority in solving the MMRSP.

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